Karol Dawid Penczek

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MSc Data Science and AI student at TU Delft and Cum Laude Computer Science graduate (TU/e), with experience building software end-to-end across REST APIs, React interfaces, Docker deployment, and resource-constrained embedded systems. Seeking a software engineering internship.

Technical Skills

- **Engineering**: Python, C/C++, JavaScript, Docker, Git, REST APIs, React.js, embedded systems (FreeRTOS)
- **Data**: SQL, data pipelines, embedding-based retrieval and similarity scoring
- **Machine Learning & AI**: PyTorch, PyTorch Geometric, TensorFlow, CNNs, GNNs, model evaluation & robustness testing, fine-tuning (LoRA), sparse autoencoders
- **Languages**: English (C1), Polish (native), German (B2/C1)

Experience

- **PeerCode (Industry Partner) — Facial Recognition Service** — Eindhoven, Netherlands
  Machine Learning / Software Engineer (six-month university–industry project team) — *September 2025 – January 2026*
  - **End-to-end delivery**: Built a facial recognition service for an industry partner, processing both images and short video clips for face detection, identification, and verification.
  - **Deployment**: Containerised the inference backend with Docker and exposed REST APIs, served through a React web interface.
  - **Data & privacy**: Assembled a GDPR-compliant synthetic identity gallery and implemented the input pipeline (face localisation, alignment, normalisation) before inference.
  - **Evaluation**: Systematically evaluated a deep face-matching model across lighting, pose, resolution, and multi-person scenarios.
  - **Security**: Designed verification and audit logging based on image hashing and embedding similarity scores for controlled, secure system usage.
- **Aster in Space — Student Team** — Eindhoven, Netherlands
  Software Engineer — *September 2024 – December 2025*
  - **AetherSense payload — live in space**: Built reliable, maintainable flight software over 16 months in a multidisciplinary team. Implemented communication, data processing, sensor scanning, and attitude determination under strict compute, power, and memory constraints, with code designed for continued team development.
  - **Design & requirements**: Compared microcontroller options and software environments, and defined system- and software-level requirements together with mechanical, electronics, and systems engineers.
  - **CubeSat project**: Selected the software environment and hardware for a 10 cm × 10 cm × 30 cm satellite and implemented software against requirements agreed across subteams.
- **ML Collective & EleutherAI** — Remote
  Member of independent ML research organisations — *October 2024 – Present*
  - **Dragonfly project (Oct – Dec 2025)**: Collaborated with an international research team (Imperial College London, UT Austin) on robustness of audio ML models, working in CLAP and AudioLDM latent spaces while preserving perceptual audio quality.
  - **Experimentation**: Selected datasets, model architectures, and evaluation setups from the academic literature; fine-tuned and evaluated the LoRAW model on the MTG-Jamendo dataset.
  - **Open source**: Contributed to the public project repository: github.com/damanimc/Dragonfly
- **Software Design Course (2IRR00), TU Eindhoven** — Eindhoven, Netherlands
  Teaching Assistant — *April 2025 – June 2025, reappointed April 2026 – June 2026*
  - Selected after finishing the course with a top grade; supported 400+ students as primary contact for a ∼60-student tutorial group, led project supervision and Q&A sessions, and graded midterm exams.

Selected Projects

- **Spatio-temporal Pedestrian Flow Prediction (GNN)** — Eindhoven, Netherlands
  Team project — individually owned the GNN model — *April 2025 – June 2025*
  - **Model**: Designed and implemented a spatio-temporal Graph Neural Network (PyTorch Geometric, GCN-based) predicting pedestrian flows in a busy train station environment.
  - **Pipeline**: Built the data transformation pipeline converting raw trajectory coordinates into time-indexed dynamic graphs (node features, edge indices, rolling prediction windows), with sequence DataLoaders and an autoregressive rollout for multi-step forecasting.
  - **Evaluation**: Added evaluation and visualisation utilities comparing predicted vs. ground-truth trajectories. Repository: github.com/karolos1444/FlowGroup1
- **Bachelor Thesis: Interpretable CNN Concept Learning via SAEs** — Eindhoven, Netherlands
  TU Eindhoven — *February 2026 – June 2026*
  - Developed a sparse-autoencoder pipeline on fine-tuned ResNet-18 representations (DeepFashion, APT-36K datasets) to study whether internal visual features decompose into interpretable concepts — relevant to model debugging and responsible AI.

Education

- **Delft University of Technology** — Delft, Netherlands
  MSc Data Science and Artificial Intelligence Technology — *September 2026 – Present (expected 2028)*
- **BlueDot Impact — Technical AI Safety Course** — Remote
  Participant, cohort-based course — *October 2026 – December 2026*
  - Studying alignment and RLHF, mechanistic interpretability, evaluations and red-teaming, AI control, and scalable oversight through technical readings, written analysis, and expert-facilitated discussions.
- **Eindhoven University of Technology** — Eindhoven, Netherlands
  BSc Computer Science and Engineering; Cum Laude — *September 2023 – August 2026*